

Fluid Mechanics and Turbo-Machinery (MEP221s)

Lecture (4)

Integral Relations for a Control Volume - Basic Laws

Energy Conservation – First Law of Thermodynamics

Linear Momentum Conservation – Newton's second law

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Course Contents

Fluid Mechanics and Turbo-Machinery (MEP221s)

1. Properties of Fluids, Density, Pressure, Viscosity, Viscous Boundary Layers
2. Fluid Statics and Pressure Measurement, Forces on Submerged Bodies,
3. Rigid Body Rotation (Rotating Fluids), Reynolds Transport Theorem on a Control Volume, Continuum Hypothesis, Streamlines, Velocity and Acceleration, Basic Laws: Continuity Equation,
4. Energy Conservation Equation, Linear Momentum Conservation
5. Angular Momentum Conservation
6. Dynamic Similarity, Similitude and Dimensional Analysis,
7. Classification of Flow Fields: Pipe Flow, Jet Flow, Wake Flow, Boundary Layer Flow, Lagrangian and Eulerian Coordinates,
8. Compressible Flow, Stagnation Properties, Mach Number and Sonic Velocity, Equations of Gas Dynamics, Flow through Nozzles, Shock Waves,
9. Classification of Turbomachines, Operation of Pumps, Series and Parallel Operation, Selection of Pumps.
10. Flow around Immersed Bodies, Drag and Lift Forces,
11. Flow in Closed Conduits, Bernoulli's Equation, Major and Minor Losses in Pipes, Laminar and Turbulent Flows,
12. Differential Analysis of Fluid Flow: Navier-Stokes Equation,

References, Books and Scientific Journals

References and Books

1. Yunus A. Cengel & John Cimbala, "Fluid Mechanics Fundamentals and Applications" Third Edition in SI units, McGraw Hill, 2014.
2. F. M. White, "Fluid Mechanics," 7th Ed., McGraw-Hill, Seventh international edition, 2012.
3. R. Munson, D. F. Young and T. H. Okishi, "Fundamentals of Fluid Mechanics," Sixth Edition, John Wiley & Sons, Inc , ISBN 0-471-24011-2, 2009.
4. John F. Douglas, "Fluid Mechanics," 5th Ed. Pearson, Printice Hall, 2005.
5. Streeter, V.L., Wylie, E.B. and Bedford, K.W., "Fluid Mechanics "Ninth edition, McGraw Hill, 2004.
6. Seppo A. Korpela, "Principles of Turbomachinery", **2011** John Wiley & Sons, Inc.
7. Rama. S.R. Gorla, Aijaz A. Khan "Turbomachinery Design and Theory," Marcel Dekker, 2003.
8. Thomas H. McCloskey, "Handbook of Turbomachinery," Hh. 8, Steam Turbines, Marcel Dekker, 2003.

Scientific Journals

- JOURNAL OF FLUID MECHANICS
- PHYSICS OF FLUIDS
- ASME Journal of Fluids Engineering
- Engineering Applications of Computational Fluid Mechanics
- COMPUTERS & FLUIDS
- JOURNAL OF TURBULENCE
- Journal of Turbomachinery – Transaction of ASME
- International Journal of Turbomachinery, Propulsion and Power

Outlines

Reynolds Transport Theorem

Objectives:

- 1. Reynolds Transport Theorem (RTT).**
- 2. Conservation of Energy (First Law of Thermodynamics).**
- 3. Conservation of Linear Momentum (Newton's Second Law).**

Approaches of the Analysis of Arbitrary Flow Problems

Experimental Analysis

It needs instrumentation and Laboratory measurement tools → expensive

Control Volume or Large-Scale (Macroscopic) Analysis (Reynolds Transport Theorem)

It is accurate for any flow distribution but is often based on average or “one-dimensional” property values at the boundaries. It always gives useful “engineering” estimates.

Differential or Small-Scale (Microscopic) Analysis

Exact (Analytical)
Solution of non-linear differential equations → very complicated and need mathematical tools.

It describes flow pattern at every point (x, y, z) in the flow field.

Numerical Modelling Analysis

The differential equations can be modelled numerically, and the flourishing field of computational fluid dynamics (CFD) can now be used to give good estimates for almost any geometry.

ANSYS-Fluent, Open FOAM and COMSOL → needs powerful digital computers.

Dimensional (Dynamic Similarity)

The dimensional analysis applies to any problem, whether analytical, numerical, or experimental. It is particularly useful to reduce the cost of experimentation

Systems versus Control Volume

All the laws of mechanics are written for a **system**, which is defined as an arbitrary quantity of mass of fixed identity. Everything external to this system is denoted by the term **surroundings**, and the system is separated from its surroundings by its **boundaries**.

System (Control Mass):

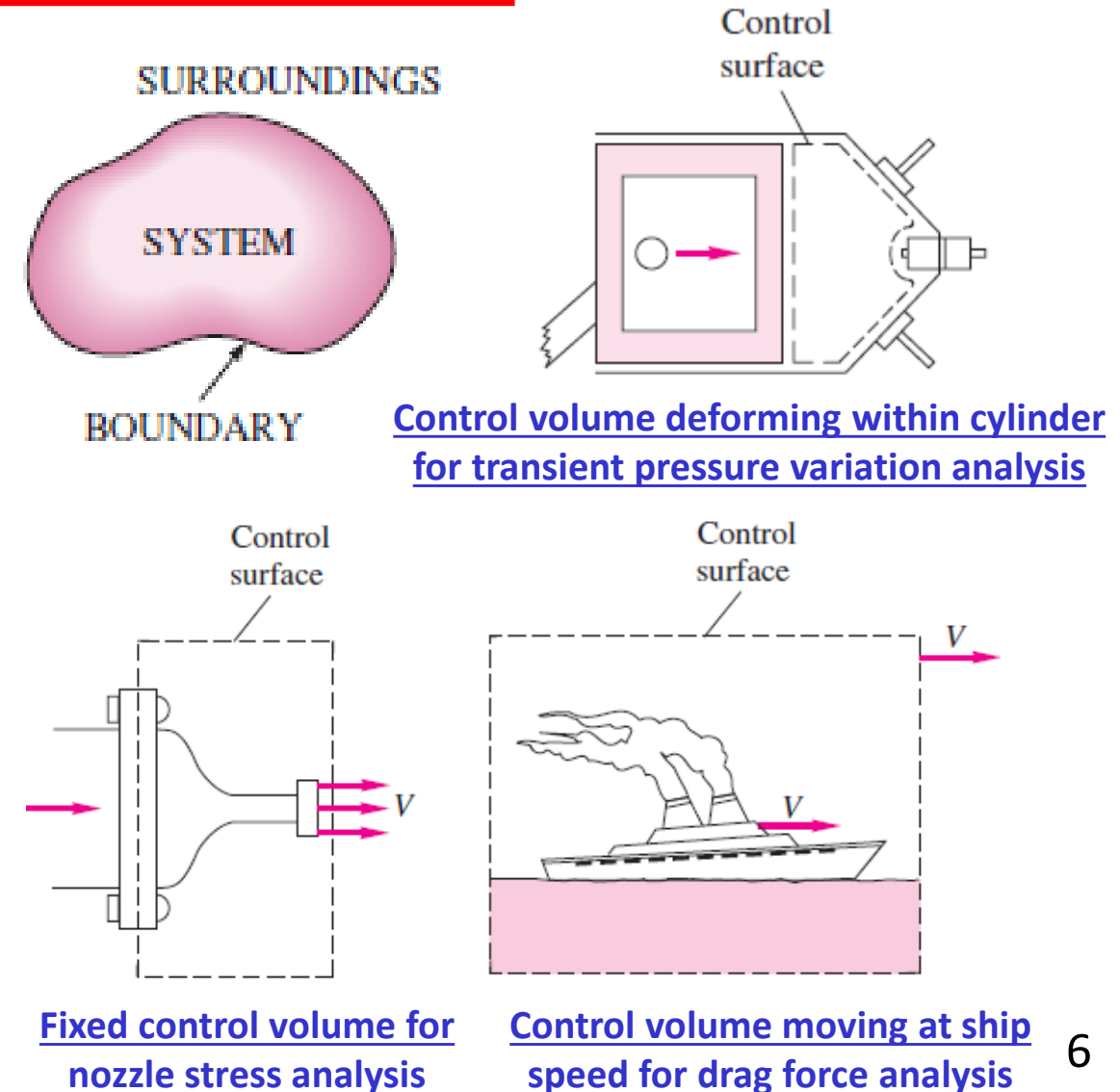
It is an arbitrary quantity of mass (collection of matter) of fixed identity (always the same fluid identity), which may move, flow, interact with its surroundings.

Control Volume:

It is a volume in space (geometric entity, independent of mass) through which fluid may flow.

Fluid Property, B:

It is any physical property associated with fluid (mass, momentum, energy).



Control Volume Approach and Reynolds Transport Theorem (RTT)

To convert a **system analysis** to a **control volume analysis**, we must convert our mathematics to apply to a **specific region** rather than to individual masses.

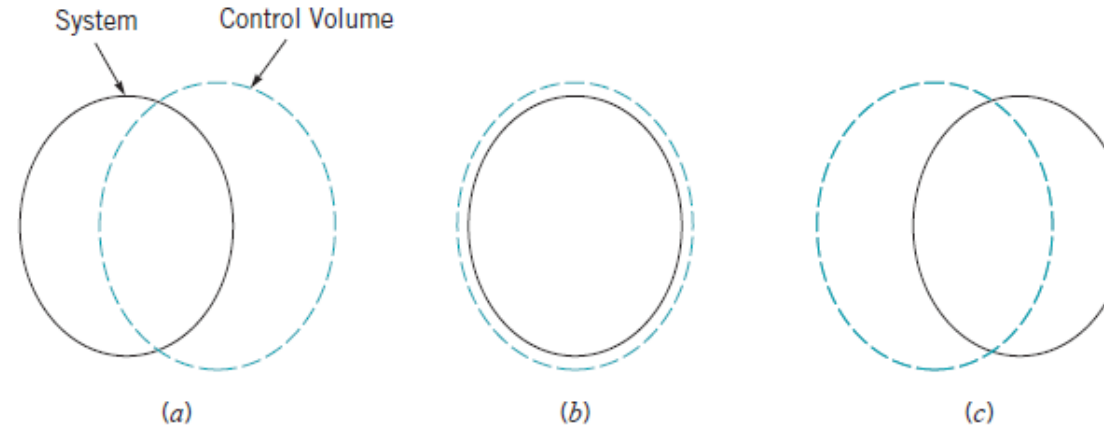
This conversion, called the **Reynolds transport theorem**, can be applied to all the basic laws.

Let B be any **extensive** property of the fluid (**energy, momentum, enthalpy, etc.**) and let

➔ $B = \text{mass, momentum, energy}$

and let β be the **intensive** value, or the amount of B per unit mass in any small element of the fluid.

➔
$$\beta = \frac{B}{m}$$



■ **FIGURE 5.1** System and control volume at three different instances of time. (a) System and control volume at time $t - \delta t$. (b) System and control volume at time t , coincident condition. (c) System and control volume at time $t + \delta t$.

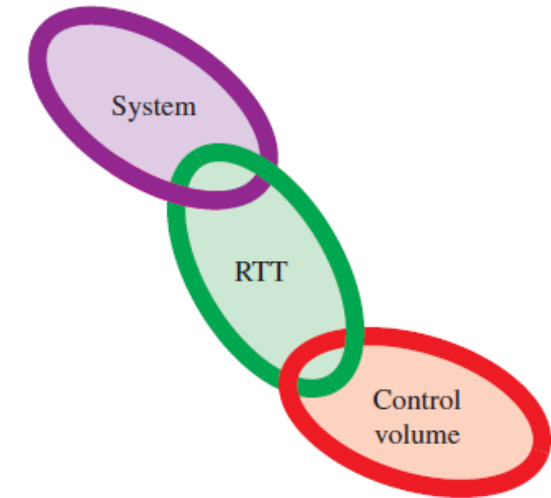


FIGURE 4-54

The *Reynolds transport theorem* (RTT) provides a link between the system approach and the control volume approach.



Reynolds Transport Theorem (RTT): System to- Control-Volume Transformation

For a system and a fixed, nondeforming control volume that are coincident at an instant of time:

Time rate of change of the system property B	=	Time rate of change (Production rate) of the property B within control volume enclosing system at any instant	+	Net outflux of the property B through the control surface (boundaries of control volume)
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$$\frac{d}{dt}(B_{\text{system}}) = \frac{d}{dt} \int_{\text{CV}} \beta \rho dV + \oint_{\text{CS}} \beta \rho \vec{V} \cdot d\vec{A}$$

CV is the control volume

CS is the control surface

β is the volume of the fluid contained in the control volume

$$\vec{V} \cdot d\vec{A} = (\vec{V} \cdot \vec{n})dA = V \cos \theta dA$$

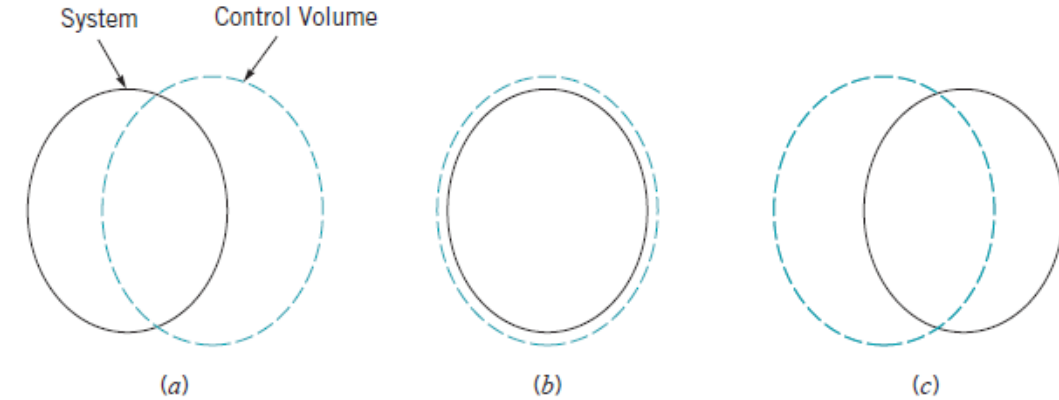


FIGURE 5.1 System and control volume at three different instances of time. (a) System and control volume at time $t - \delta t$. (b) System and control volume at time t , coincident condition. (c) System and control volume at time $t + \delta t$.

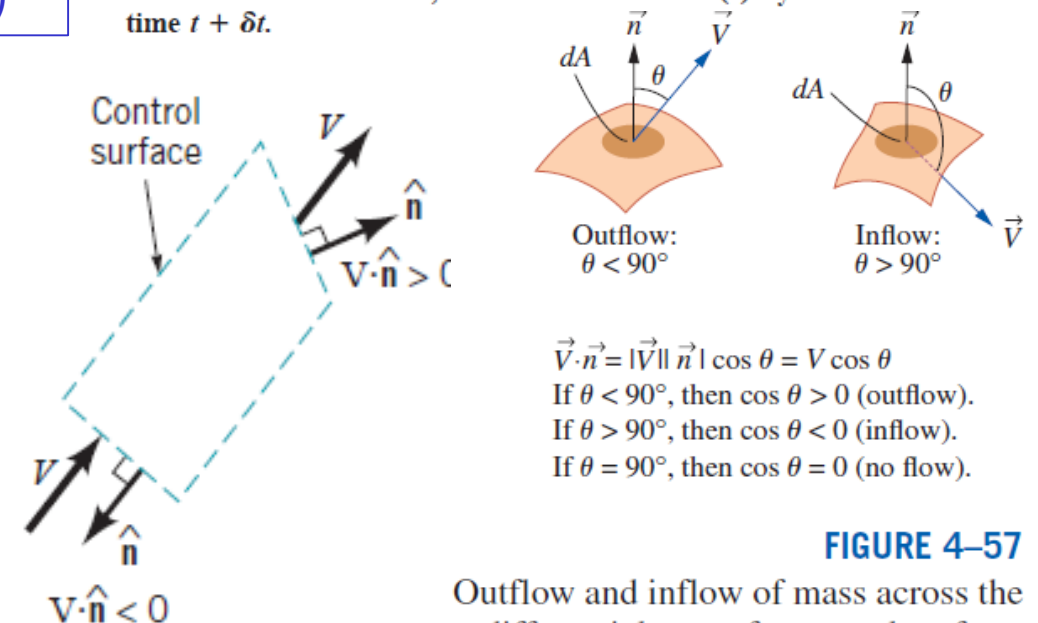


FIGURE 4-57

Outflow and inflow of mass across the differential area of a control surface.

General Integral Conservation of Energy – First Law of Thermodynamics

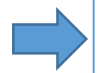


$$\frac{d}{dt}(B_{\text{system}}) = \frac{d}{dt} \int_{\text{CV}} \beta \rho dV + \oint_{\text{CS}} \beta \rho \vec{V} \cdot d\vec{A}$$

$$B = E$$

$$\beta = \frac{E}{m} = e$$

the time rate of increase of the total stored energy of the system = the time rate of increase of the total stored energy of the contents of the control volume + the net rate of flow of the total stored energy out of the control volume through the control surface



$$\frac{dQ}{dt} - \frac{dW}{dt} = \left(\frac{dE}{dt} \right)_{\text{system}} = \frac{d}{dt} \int_{\text{CV}} e \rho dV + \oint_{\text{CS}} e \rho (\vec{V} \cdot \vec{n}) dA$$

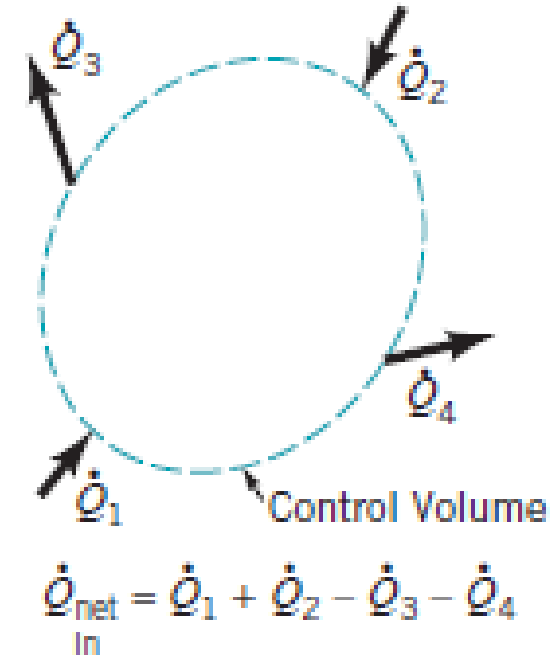


$$\vec{V} \cdot d\vec{A} = (\vec{V} \cdot \vec{n}) dA = V \cos \theta dA$$

$$e = u + \frac{V^2}{2} + gz$$

$$\dot{Q} = \dot{Q}_{\text{in}} - \dot{Q}_{\text{out}}$$

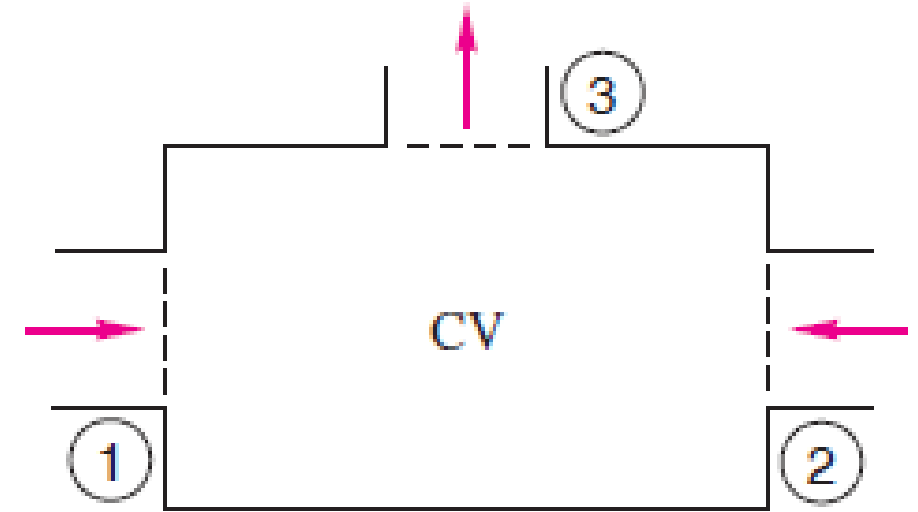
$$\dot{W} = \dot{W}_{\text{in}} - \dot{W}_{\text{out}}$$



$\dot{Q}$: net rate of heat transfer into the system [W]
 $\dot{W}$: net rate of work transfer into the system [W]
 h: enthalpy [J/kg]
 u: internal energy [J/kg]
 V: velocity [m/s]
 z: potential head [m]
 g: gravitational acceleration [m/s²]

Solved Example (1)

A fixed control volume has three one-dimensional boundary sections, as shown in the attached Fig. The flow within the control volume is steady. The flow properties at each section are tabulated below. Find the rate of change of energy of the system that occupies the control volume at this instant. Also, check the balance of mass.



Section	Type	ρ , kg/m ³	V , m/s	A , m ²	e , J/kg
1	Inlet	800	5.0	2.0	300
2	Inlet	800	8.0	3.0	100
3	Outlet	800	17.0	2.0	150

Solved Example (1)

The rate of change of energy of the system that occupies the control volume at this instant:

$$\Rightarrow \left(\frac{dE}{dt} \right)_{\text{system}} = \frac{d}{dt} \int_{\text{CV}} e \rho dV + \oint_{\text{CS}} e \rho (\vec{V} \cdot \vec{n}) dA$$

$$\left(\frac{dE}{dt} \right)_{\text{system}} = \cancel{\frac{d}{dt} \int_{\text{CV}} e \rho dV} + (e_3 \dot{m}_3 - e_1 \dot{m}_1 - e_2 \dot{m}_2)$$

Steady state one-dimensional
boundary sections

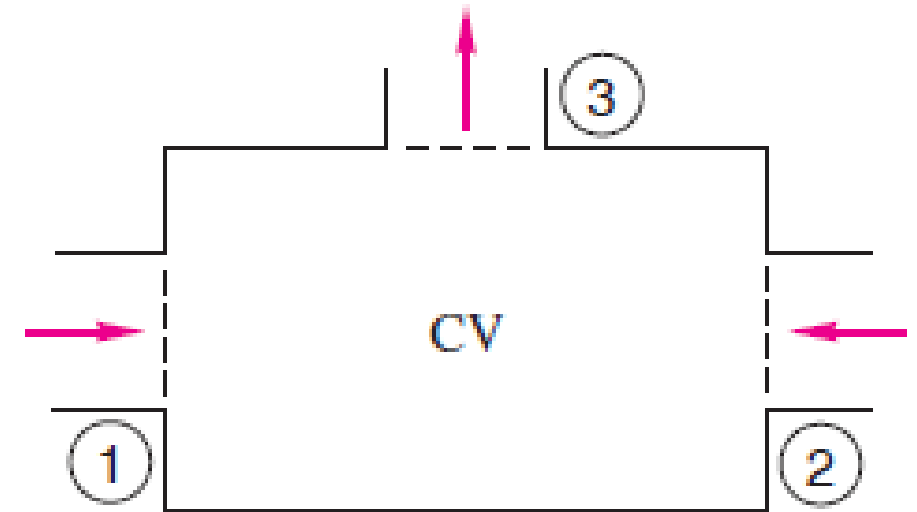
$$\left(\frac{dE}{dt} \right)_{\text{system}} = 0 + (e_3 \rho_3 A_3 V_3 - e_1 \rho_1 A_1 V_1 - e_2 \rho_2 A_2 V_2)$$

Substitute:



$$\left(\frac{dE}{dt} \right)_{\text{system}} = -240000 \frac{\text{J}}{\text{s}}$$

Thus the system is losing energy



Solved Example (1)

Check the balance of mass:



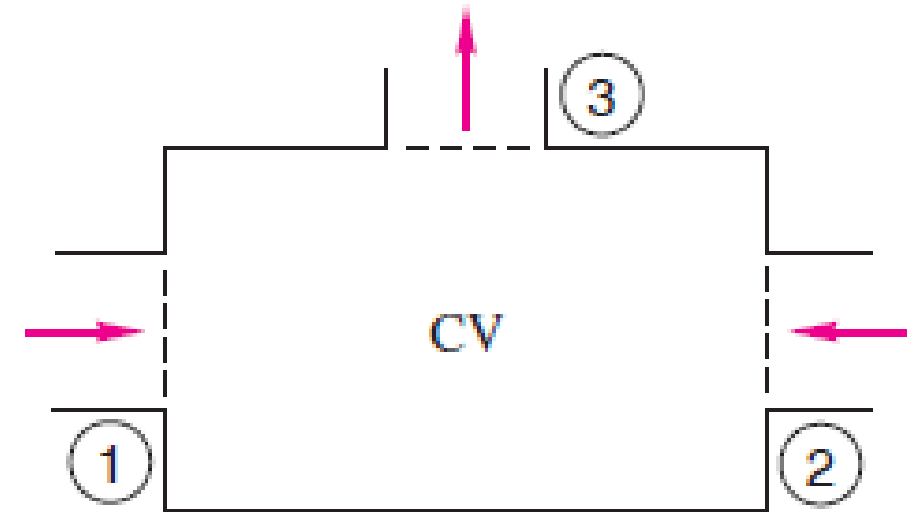
$$\left(\frac{dm}{dt}\right)_{system} = \frac{d}{dt} \int_{CV} \rho \, dV + \oint_{CS} \rho (\vec{V} \cdot \vec{n}) \, dA$$

$$\left(\frac{dE}{dt}\right)_{system} = \frac{d}{dt} \int_{CV} e \rho \, dV + (\dot{m}_3 - \dot{m}_1 - \dot{m}_2)$$

Steady state

one-dimensional
boundary sections

$$\left(\frac{dm}{dt}\right)_{system} = 0 + (\rho_3 A_3 V_3 - \rho_1 A_1 V_1 - \rho_2 A_2 V_2)$$



Substitute:



$$\left(\frac{dm}{dt}\right)_{system} = 0 \frac{\text{kg}}{\text{s}}$$

Thus the system mass does not change, which correctly expresses the law of conservation of system mass,

General Integral Conservation of Energy – First Law of Thermodynamics

$$\dot{W} = \dot{W}_{shaft} + \dot{W}_{press} + \dot{W}_{viscous}$$

$\dot{W}_{shaft}$: rate of shaft work [W]

$\dot{W}_{press}$: rate of pressure (displacement) work [W]

$\dot{W}_{viscous}$: rate of viscous (shear) work [W]

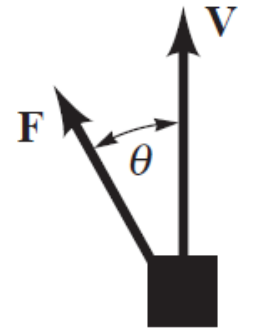
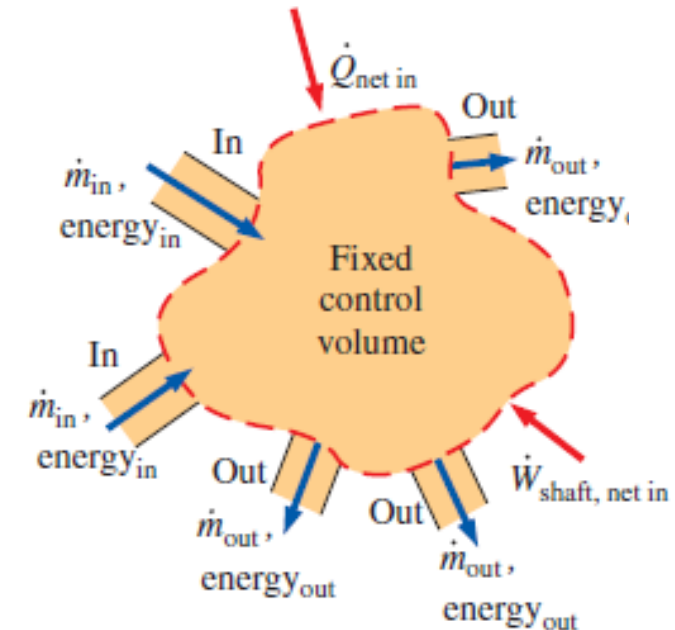
$$\dot{W}_{press} = \oint_{CS} p (\vec{V} \cdot \vec{n}) dA$$

The shear work term $\dot{W}_{viscous}$ is rarely important

$$\dot{Q} - \dot{W}_{shaft} - \dot{W}_{press} - \dot{W}_{viscous} = \frac{d}{dt} \int_{CV} e \rho dV + \oint_{CS} e \rho (\vec{V} \cdot \vec{n}) dA$$

$$\dot{Q} - \dot{W}_{shaft} - \dot{W}_{viscous} = \frac{d}{dt} \int_{CV} e \rho dV + \oint_{CS} \left(e + \frac{p}{\rho} \right) \rho (\vec{V} \cdot \vec{n}) dA$$

$$\dot{Q} - \dot{W}_{shaft} - \dot{W}_{viscous} = \frac{d}{dt} \int_{CV} \left(u + \frac{V^2}{2} + gz \right) \rho dV + \oint_{CS} \left(u + \frac{V^2}{2} + gz + \frac{p}{\rho} \right) \rho (\vec{V} \cdot \vec{n}) dA$$



$$\dot{W} = \mathbf{F} \cdot \mathbf{V} = FV \cos \theta$$

**General Integral
Energy Conservation
Equation**

General Integral Conservation of Energy – First Law of Thermodynamics

General Integral Energy Conservation Equation:

$$\dot{Q} - \dot{W}_{shaft} - \dot{W}_{viscous} = \frac{d}{dt} \int_{CV} \left(u + \frac{V^2}{2} + gz \right) \rho \, dV + \oint_{CS} \left(h + \frac{V^2}{2} + gz \right) \rho (\vec{V} \cdot \vec{n}) \, dA$$

$$\frac{p}{\rho} = \text{flow work}$$

$$h = u + \frac{p}{\rho}$$

For One- Dimensional inlets and Outlets:

$$\dot{Q} - \dot{W}_{shaft} - \dot{W}_{viscous} = \frac{d}{dt} \int_{CV} \left(u + \frac{V^2}{2} + gz \right) \rho \, dV + \sum_i^{CS} \left(\dot{m}_i \left(h_i + \frac{V_i^2}{2} + gz_i \right) \right)_{out} - \sum_i^{CS} \left(\dot{m}_i \left(h_i + \frac{V_i^2}{2} + gz_i \right) \right)_{in}$$

For steady-state, steady-flow (SSSF) - One- Dimensional inlets and Outlets:

$$\dot{Q} - \dot{W}_{shaft} - \dot{W}_{viscous} = \sum_i^{CS} \left(\dot{m}_i \left(h_i + \frac{V_i^2}{2} + gz_i \right) \right)_{out} - \sum_i^{CS} \left(\dot{m}_i \left(h_i + \frac{V_i^2}{2} + gz_i \right) \right)_{in}$$

Note:

The shear work term $\dot{W}_{viscous}$ is rarely important

For steady-state, steady-flow of single stream of fluid flow with no heat or work interactions:

$$h_{in} + \frac{V_{in}^2}{2} + gz_{in} = h_{out} + \frac{V_{out}^2}{2} + gz_{out}$$

$$\dot{Q} = 0$$

$$\dot{W}_{shaft} = 0$$

$$\dot{W}_{viscous} = 0$$

Steady-State, Steady-Flow (SSSF) Energy Equation, One- Dimensional Inlets and Outlets

For One inlet and one outlet:

$$\Rightarrow \dot{Q} - \dot{W}_{shaft} - \dot{W}_{viscous} = \dot{m}_{in} \left(h_{in} + \frac{V_{in}^2}{2} + gz_{in} \right) - \dot{m}_{out} \left(h_{out} + \frac{V_{out}^2}{2} + gz_{out} \right) \quad \dot{m}_{in} = \dot{m}_{out} = \dot{m}$$

Dividing by $\dot{m}$:

$$\Rightarrow \left(h_{in} + \frac{V_{in}^2}{2} + gz_{in} \right) = \left(h_{out} + \frac{V_{out}^2}{2} + gz_{out} \right) - q + w_{shear} + w_{viscous} \quad h = u + \frac{p}{\rho}$$

$$q = \frac{\dot{Q}}{\dot{m}} \left[\frac{J}{kg} \right]$$

$$w_{shaft} = \frac{\dot{W}_{shaft}}{\dot{m}} \left[\frac{J}{kg} \right]$$

$$w_{viscous} = \frac{\dot{W}_{viscous}}{\dot{m}} \left[\frac{J}{kg} \right]$$

Dividing by g :

Let section 1 be the inlet and section 2 the outlet:

$$\Rightarrow \left(\frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z_1 \right) + \frac{u_1}{g} = \left(\frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2 \right) + \frac{u_2}{g} - h_q + h_{shaft} + h_{viscous}$$

$$h_q = \frac{q}{g} [m]$$

$$h_{shaft} = \frac{w_{shear}}{g} [m]$$

$$h_{viscous} = \frac{w_{viscous}}{g} [m]$$

Note:

The shear work term $\dot{W}_{viscous}$ is rarely important

Head Form

Ideal Bernoulli Equation

$$\Rightarrow \left(\frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z_1 \right) = \left(\frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2 \right) + \left(\frac{u_2}{g} - \frac{u_1}{g} \right) - h_q + h_{shaft} + h_{viscous}$$

Head Form

$$\left(\frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z_1 \right) = \left(\frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2 \right) + h_{friction} + h_{shaft} + h_{viscous}$$

$$h_{friction} = \left(\frac{u_2}{g} - \frac{u_1}{g} \right) - h_q$$

$h_{friction}$: energy head loss due to friction [m]

$$h_{shaft} = h_{turbine} - h_{pump}$$

For no Friction (Frictionless Flow):

$$\Rightarrow h_{friction} = 0$$

If there is no pump or turbine work, no friction:

Ideal Bernoulli Equation

$$\left(\frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z_1 \right) = \left(\frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2 \right)$$

Conditions of Ideal Bernoulli Equation:

1. Frictionless flow
2. No shaft work (no pump, no turbine)
3. No heat transfer
4. Steady-state, steady flow
5. Incompressible flow

Ideal Bernoulli Equation

The Bernoulli equation was first stated in words by the Swiss mathematician **Daniel Bernoulli** (1700–1782) in a text written in 1738 when he was working in St. Petersburg, Russia.

It was later derived in equation form by his associate **Leonhard Euler** in 1755.

We recognize $V^2/2$ as **kinetic energy**, gz as **potential energy**, and p/ρ as **flow energy**, all per unit mass.

Therefore, the Bernoulli equation can be viewed as an expression of mechanical energy balance and can be stated as follows:

The sum of the kinetic, potential, and flow energies of a fluid particle is constant along a streamline during steady flow when the compressibility and frictional effects are negligible.

$$\Rightarrow \frac{p_1}{\rho} + \frac{1}{2} V_1^2 + g z_1 = \frac{p_2}{\rho} + \frac{1}{2} V_2^2 + g z_2$$



Daniel Bernoulli (1700–1782)

Ideal Bernoulli Equation

$$\left(\frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z_1 \right) = \left(\frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2 \right)$$

The **ideal Bernoulli equation** is a relation between **pressure**, **velocity** and **elevation** in a flow field, and is valid in regions of **steady**, **incompressible** flow where net frictional forces are negligible.

Proof:

The velocity **V** of a fluid particle is a function of space **s** and time **t**. Therefore:

$$V = V(s, t)$$



$$dV = \frac{\partial V}{\partial s} ds + \frac{\partial V}{\partial t} dt$$

In steady flow:

$$\frac{\partial V}{\partial t} = 0$$

thus

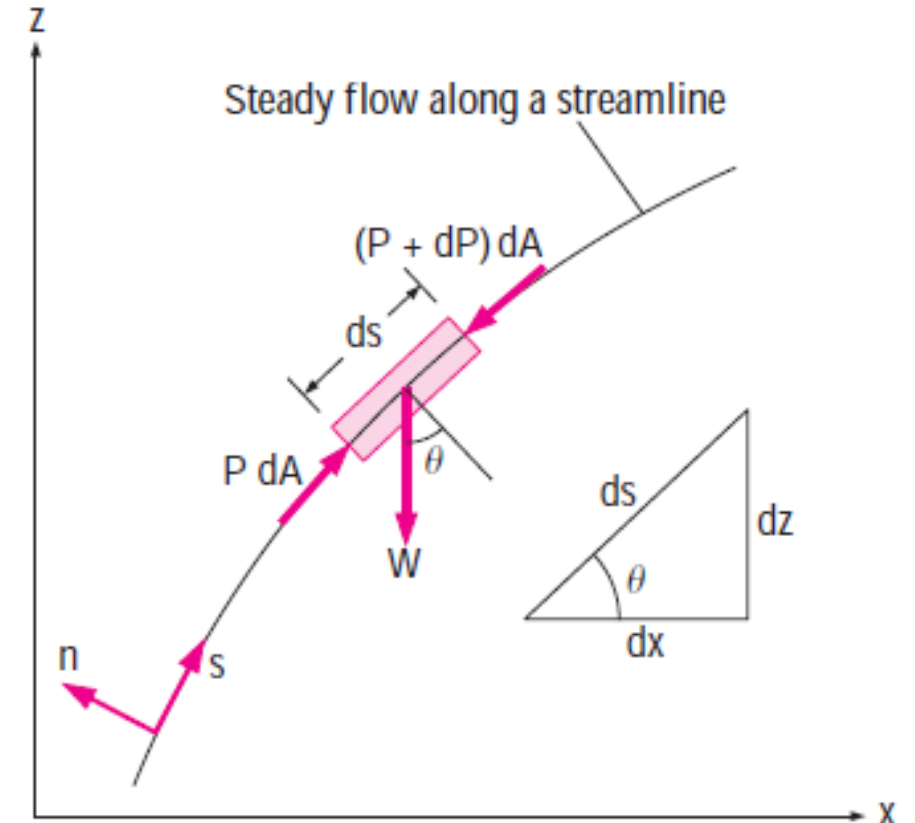


$$V = V(s)$$

and the acceleration in the s-direction becomes:



$$a_s = \frac{dV}{dt} = \frac{\partial V}{\partial s} \frac{ds}{dt} = \frac{\partial V}{\partial s} V = V \frac{dV}{ds}$$



Ideal Bernoulli Equation

From Newton's second law:

$$\Rightarrow \sum F_s = ma_s \quad \Rightarrow \quad p dA - (p + dp) dA - W \sin \theta = m V \frac{dV}{ds}$$

$$\Rightarrow \quad -dp dA - \rho g dA ds \frac{dz}{ds} = \rho dA ds V \frac{dV}{ds}$$

Euler's Equation

$$\Rightarrow \quad -dp - \rho g dz = \rho V dV \quad \Rightarrow \quad \frac{dp}{\rho} + \frac{1}{2} dV^2 + g dz = 0$$

$$\int \left(\frac{dp}{\rho} + \frac{1}{2} dV^2 + g dz \right) = \text{constant}$$

Along a streamline:

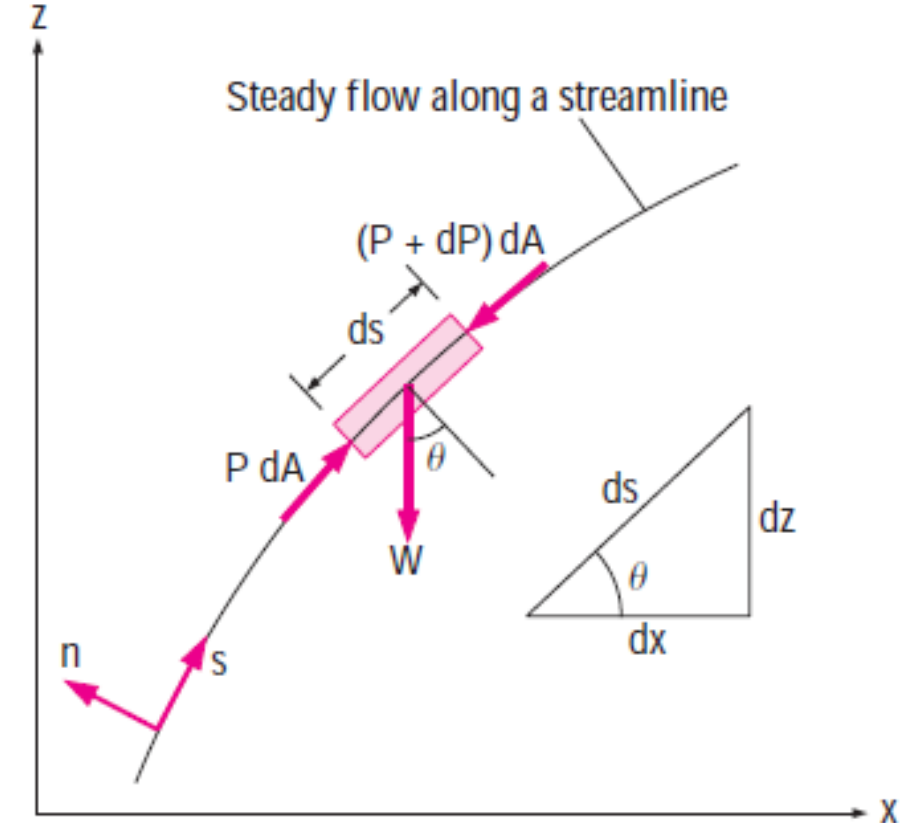
For steady-incompressible flow:

$$\Rightarrow \quad \frac{p}{\rho} + \frac{1}{2} V^2 + g z = \text{constant}$$

Along a streamline:

For steady-incompressible flow along streamline, between two points 1 & 2:

$$\Rightarrow \quad \frac{p_1}{\rho} + \frac{1}{2} V_1^2 + g z_1 = \frac{p_2}{\rho} + \frac{1}{2} V_2^2 + g z_2$$



Ideal Bernoulli Equation

p is the **static pressure** (it does not incorporate any dynamic effects); it represents the actual thermodynamic pressure of the fluid. This is the same as the pressure used in thermodynamics and property tables.

$\rho V^2/2$ is the **dynamic pressure**; it represents the pressure rise when the fluid in motion is brought to a stop isentropically.

$\rho g z$ is the **hydrostatic pressure (potential head)**, which is not pressure in a real sense since its value depends on the reference level selected; it accounts for the elevation effects of fluid weight on pressure.

along a streamline

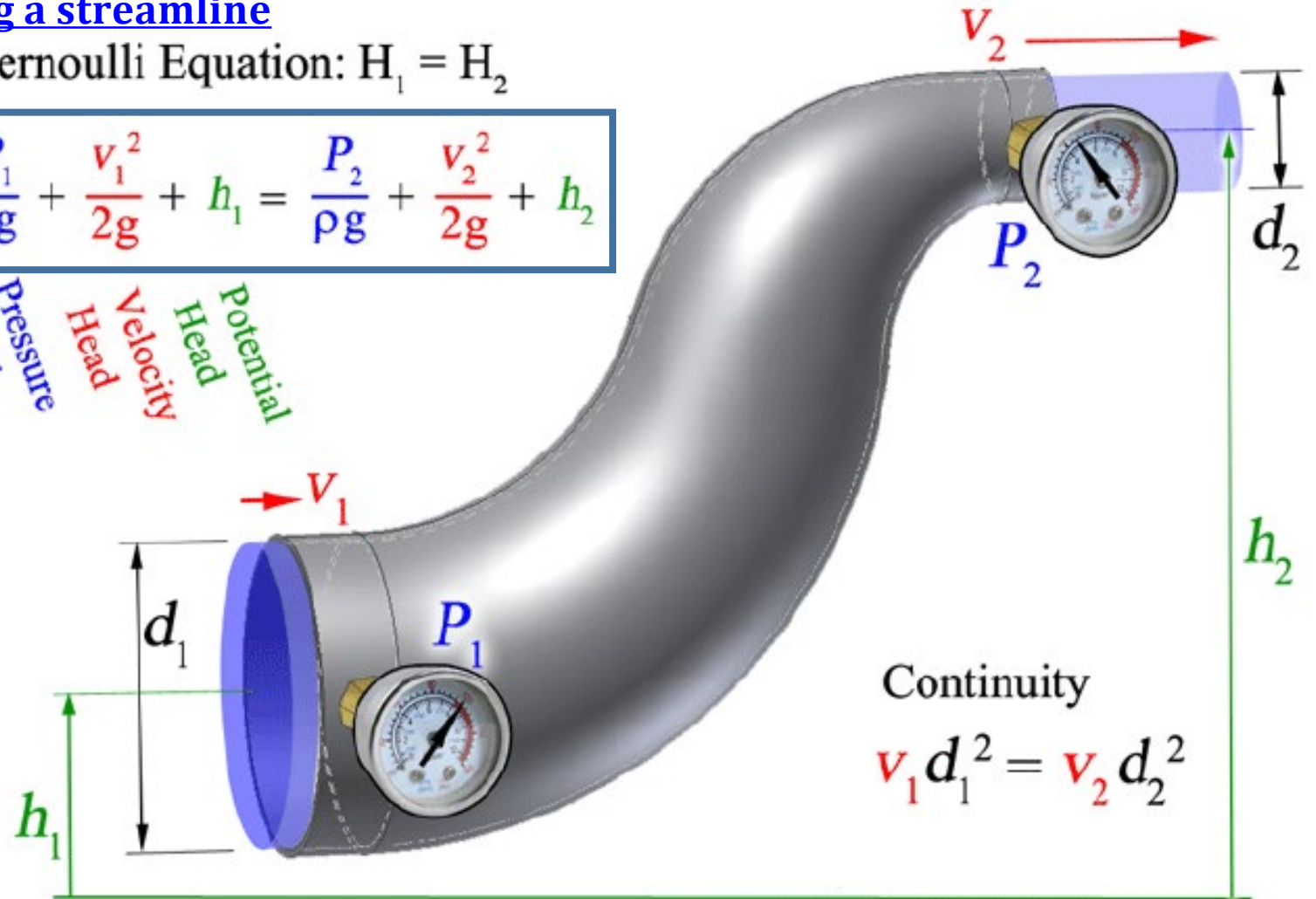
Bernoulli Equation: $H_1 = H_2$

$$\frac{P_1}{\rho g} + \frac{v_1^2}{2g} + h_1 = \frac{P_2}{\rho g} + \frac{v_2^2}{2g} + h_2$$

Pressure Head

Velocity Head

Potential Head



Continuity

$$V_1 d_1^2 = V_2 d_2^2$$

The Linear Momentum Equation – Newton's Second Law

Reynolds Transport Theorem:

$$\frac{d}{dt}(B_{\text{system}}) = \frac{d}{dt} \int_{\text{CV}} \beta \rho dV + \oint_{\text{CS}} \beta \rho \vec{V} \cdot d\vec{A}$$

$$B = mV$$

$$\beta = V$$

$$\frac{d}{dt}(m\vec{V})_{\text{system}} = \frac{d}{dt} \int_{\text{CV}} \vec{V} \rho dV + \oint_{\text{CS}} \vec{V} \rho (\vec{V} \cdot \vec{n}) dA$$

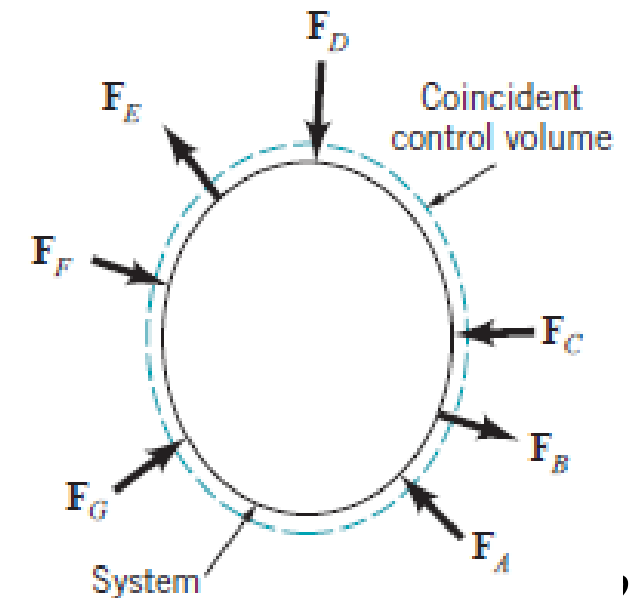
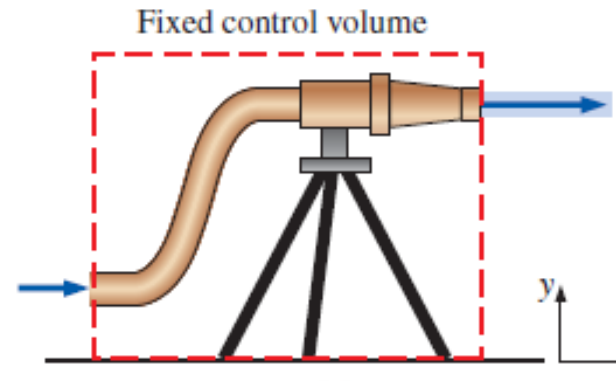
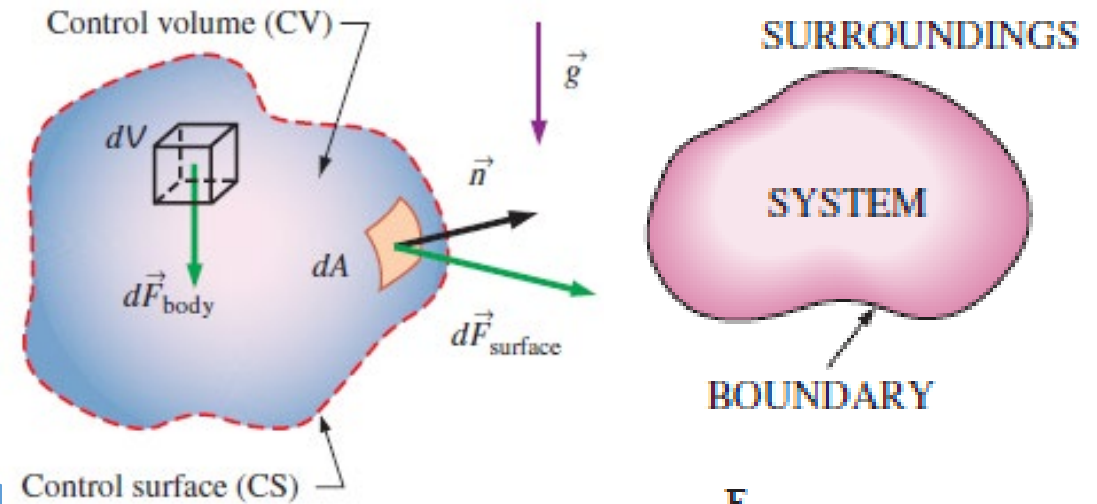
Newton Second Law:

$$\frac{d}{dt}(m\vec{V})_{\text{system}} = \sum \vec{F}$$

$$\vec{V} \cdot d\vec{A} = (\vec{V} \cdot \vec{n}) dA = V \cos \theta dA$$

$\sum \vec{F}$ is the vector sum of all forces acting on the control-volume material considered as a free body.

$$\sum \vec{F} = \frac{d}{dt} \int_{\text{CV}} \vec{V} \rho dV + \underbrace{\oint_{\text{CS}} \vec{V} \rho (\vec{V} \cdot \vec{n}) dA}_{\text{Momentum flow rate}}$$



The Linear Momentum Equation – Newton's Second Law

$$\sum \vec{F} = \frac{d}{dt} \int_{CV} \vec{V} \rho dV + \underbrace{\sum_i^{CS} (\dot{m}_i \vec{V})_{out}}_{\text{Outlet momentum flow rate}} - \underbrace{\sum_i^{CS} (\dot{m}_i \vec{V})_{in}}_{\text{Inlet momentum flow rate}}$$

for steady flow:

$$\frac{d}{dt} \int_{CV} \vec{V} \rho dV = 0$$

Linear momentum conservation
in x-direction:

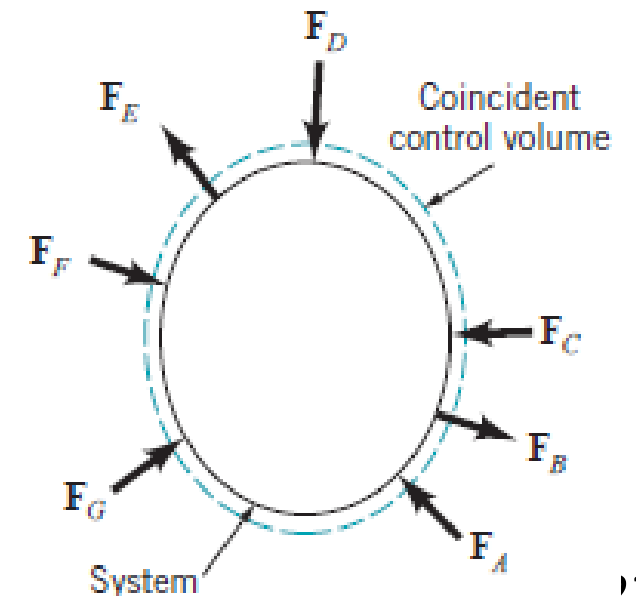
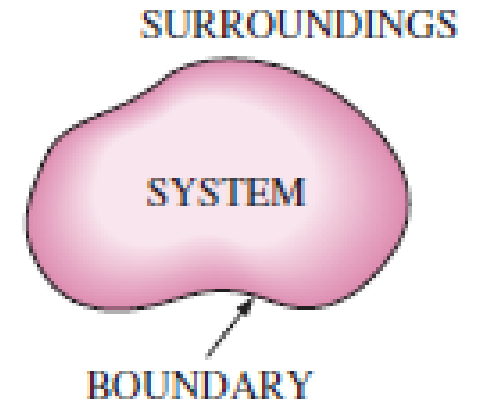
$$\sum F_x = \frac{d}{dt} \int_{CV} u \rho dV + \sum_i^{CS} (\dot{m}_i u)_{out} - \sum_i^{CS} (\dot{m}_i u)_{in}$$

Linear momentum conservation
in y-direction:

$$\sum F_y = \frac{d}{dt} \int_{CV} v \rho dV + \sum_i^{CS} (\dot{m}_i v)_{out} - \sum_i^{CS} (\dot{m}_i v)_{in}$$

Linear momentum conservation
in z-direction:

$$\sum F_z = \frac{d}{dt} \int_{CV} w \rho dV + \sum_i^{CS} (\dot{m}_i w)_{out} - \sum_i^{CS} (\dot{m}_i w)_{in}$$



The Linear Momentum Equation – Newton's Second Law

Linear momentum conservation:

$$\sum \mathbf{F}_j = \frac{d}{dt} \int_{CV} \mathbf{V}_j \rho dV + \sum_i^{CS} (\dot{m}_i \mathbf{V}_{ij})_{out} - \sum_i^{CS} (\dot{m}_i \mathbf{V}_{ij})_{in}$$

where $i = 1, 2, 3, 4, \dots$ etc and $j = x, y, z$ in Cartesian coordinates

For steady flow:

$$\sum \mathbf{F}_j = \sum_i^{CS} (\dot{m}_i \mathbf{V}_{ij})_{out} - \sum_i^{CS} (\dot{m}_i \mathbf{V}_{ij})_{in}$$

Control
volume

Positive
directions

Inputs and
outputs

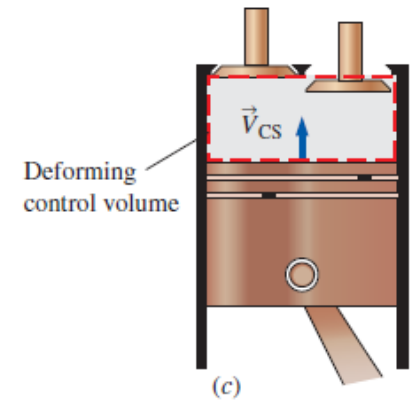
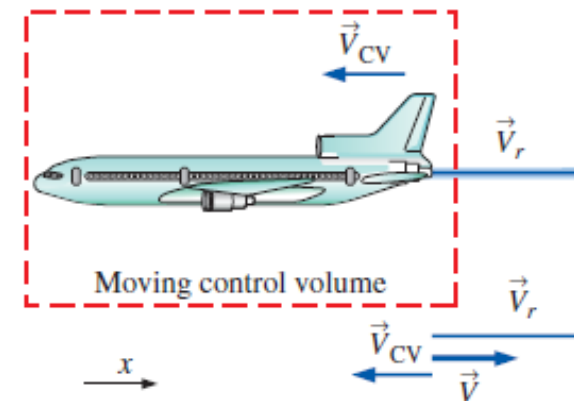
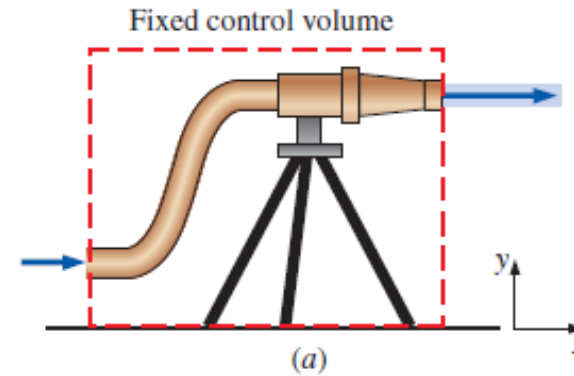
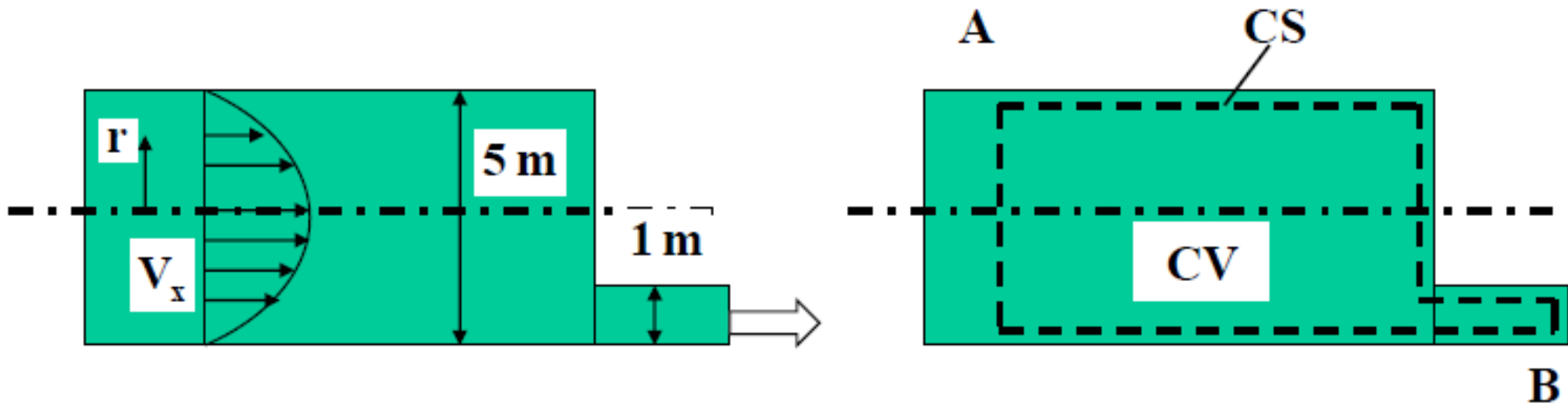


FIGURE 6-4

Examples of (a) fixed, (b) moving, and (c) deforming control volumes.

Solved Example (2)

Water is flowing through a large pipe of diameter 5 m. The velocity of water relative to the tank is given as: $V_x = 6.25 - r^2$. What is the **average velocity** of the water flowing through the smaller pipe having an inside diameter of 1 m?



Solved Example (2)

Conservation of mass:

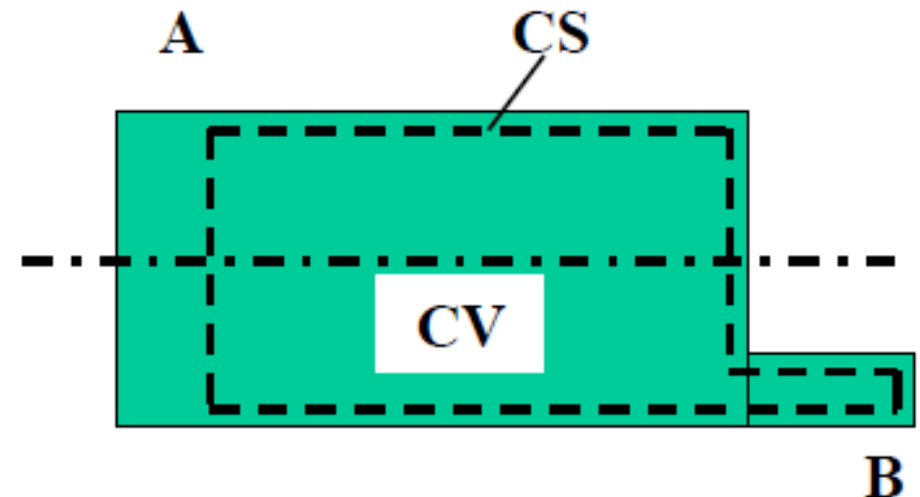
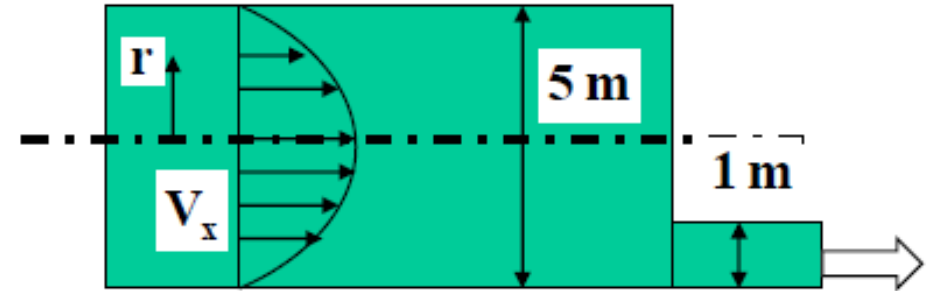
$$\Rightarrow \frac{d}{dt}(m_{\text{system}}) = 0 = \frac{d}{dt} \int_{\text{CV}} \rho dV + \oint_{\text{CS}} \rho \vec{V} \cdot \vec{dA}$$

Given: Steady state:

$$\Rightarrow \oint_{\text{inCS}}^{\text{outCS}} \rho \vec{V} \cdot \vec{dA} = 0 \quad \Rightarrow \quad \rho V_B \left(\frac{\pi}{4} D_B^2 \right) - \int_0^{2.5} \rho V_x 2\pi r dr = 0$$

$$\Rightarrow V_B \left(\frac{\pi}{4} D_B^2 \right) = \int_0^{2.5} (6.25 - r^2) 2\pi r dr$$

$$V_B \left(\frac{\pi}{4} 1^2 \right) = 2\pi \left(6.25 \frac{r^2}{2} - \frac{r^4}{4} \right) \Big|_0^{2.5} \quad \Rightarrow \quad V_B = 78.1 \text{ m/s}$$

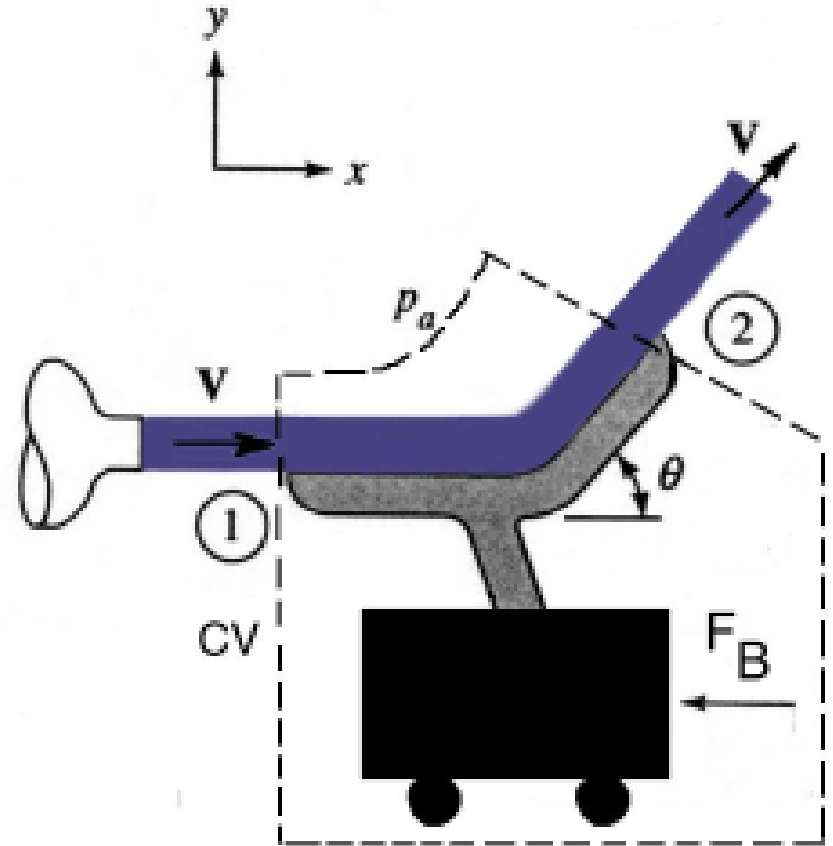


Solved Example (3)

A water jet 4 cm in diameter with a velocity of 7 m/s is directed to a stationary turning vane with $\theta = 40^\circ$. Determine the force, F_B , necessary to hold the vane stationary.

Assignment

What happens if θ is zero?. Draw the variation of the force F_B with θ varying from 0 to 180 and comment on your results.



Solved Example (3)

Linear momentum conservation in x-direction:

$$\rightarrow \sum F_x = \frac{d}{dt} \int_{CV} u \rho dV + \sum_i^{CS} (\dot{m}_i u)_{out} - \sum_i^{CS} (\dot{m}_i u)_{in}$$

Since the flow is **steady** and the CV is stationary, the time rate of change of momentum within the CV is zero. Also, with uniform velocity at each inlet and exit, and a constant flow rate, the momentum equation becomes:

$$\vec{F}_B = \dot{m}(\vec{V}_o - \vec{V}_i)$$

$$\rightarrow -F_B = \rho V \left(\frac{\pi}{4} 0.04^2 \right) (V \cos 40^\circ - V)$$

$$\rightarrow F_B = 1000.7^2 \cdot \left(\frac{\pi}{4} 0.04^2 \right) (1 - \cos 40^\circ) = 14.4 \text{ N}$$

